

Algebra 1

Unit 2

Lesson 4 Homework

Name _____

Date _____

Period _____

1. True or False? The polynomial $-7xy^4 + 4x^5y^2 - 8.6x^3y^3$ is written in standard form.

2. Write the polynomial $-7xy^4 + 4x^5y^2 - 8.6x^3y^3$ in standard form.

For questions 3-4, rewrite each polynomial in standard form, then identify the features of the polynomial.

3. $2.5y^3 + 7y - 8\frac{7}{8}y^5$

Standard Form _____

Number of Terms _____

Degree of the polynomial _____

Leading Term _____

Leading Coefficient _____

4. $-\frac{3}{4}a^4b^2 - 16ab^3 + ab^6$

Standard Form _____

Number of Terms _____

Degree of the polynomial _____

Leading Term _____

Leading Coefficient _____

5. Create a polynomial that has 4 terms, a degree of 6, a leading coefficient of -1 , and a constant term of -5.5 .
6. Samantha found the product of $-2g^3(\frac{1}{4}g^8h^3 + 5g^2h^6 - g^9)$ to be $-\frac{1}{2}g^{24}h^3 - 10g^6h^6 - 2g^{27}$. Do you agree or disagree with her? Explain.
7. Find the product of $-2g^3(\frac{1}{4}g^8h^3 + 5g^2h^6 - g^9)$ in standard form.

Find the product of each by distributing the monomial to the polynomial. Express the product in standard form.

8. $-5x^2(-3xy + 2x^7 - 7x^3)$
9. $(-\frac{5}{8}c^4d + 10c^5d^3 - cd^2)(\frac{3}{5}c^2)$
10. $-4.2a^3(2b^5 - 1.5a^2b - 9)$